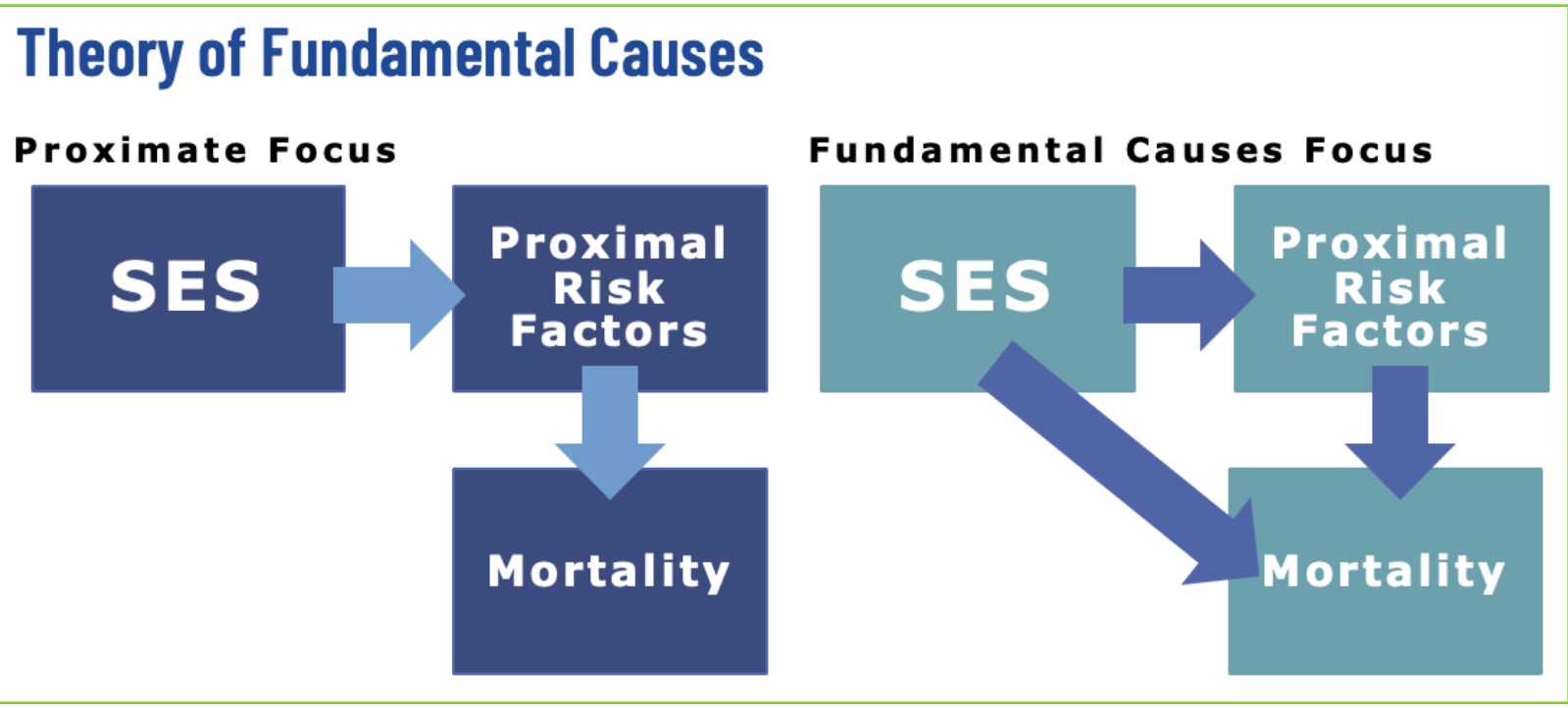


Understanding the Relationship Between Persistent Predictors and Persistent COVID: An analysis of Long-term COVID Effects, Chronic Health Conditions, and Sociodemographic Characteristics of US Adults

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Background

- The World Health Organization (WHO) reports **103 million total COVID cases in the United States since 2020**, demonstrating a large and widespread disease burden impacting various people.
- Recent epidemiological studies report that **long-term COVID effects “negatively impact functioning, and quality of life and mental health**, with cognitive and physical impairments being prevalent.”
- Sociodemographic analyses consistently show **disparities in the prevalence of chronic health conditions and multimorbidity by race and income**.
- When predicting the prevalence and severity of long-term COVID impairments, **previous research evaluated factors like multimorbidity, smoking status, and depression**.
- The Theory of Fundamental Causes argues that **Socioeconomic Status (SES) shapes multiple disease outcomes through its influence on proximal risk factors** like smoking, diet, and access to care.



- The present study aims to understand the relationship between **chronic health conditions and long-term COVID effects within adults in the United States**, with specific consideration of SES as a fundamental cause influencing the primary predictor of interest.

Methods

- Data from core, emerging core, and optional modules of the 2023 Behavioral Risk Factor Surveillance System (BRFSS) informed this analysis.

Primary Predictor: Chronic Multimorbidity

- A categorical scale totaling the presence of 10 chronic health conditions: heart attack, angina/coronary heart disease, stroke, asthma, cancer, C.O.P.D/emphysema/chronic bronchitis, depression, Kidney disease, arthritis/rheumatoid arthritis/gout/lupus/fibromyalgia, and diabetes/pre-diabetes/ borderline diabetes

Demographic Predictors

- Sex, race, education, age, and household annual income

Additional Predictors

- Health insurance status, health care affordability, COVID-19 vaccination status, and smoking status

Data Analysis

Univariate analysis -> produced frequencies of all study variables of interest.

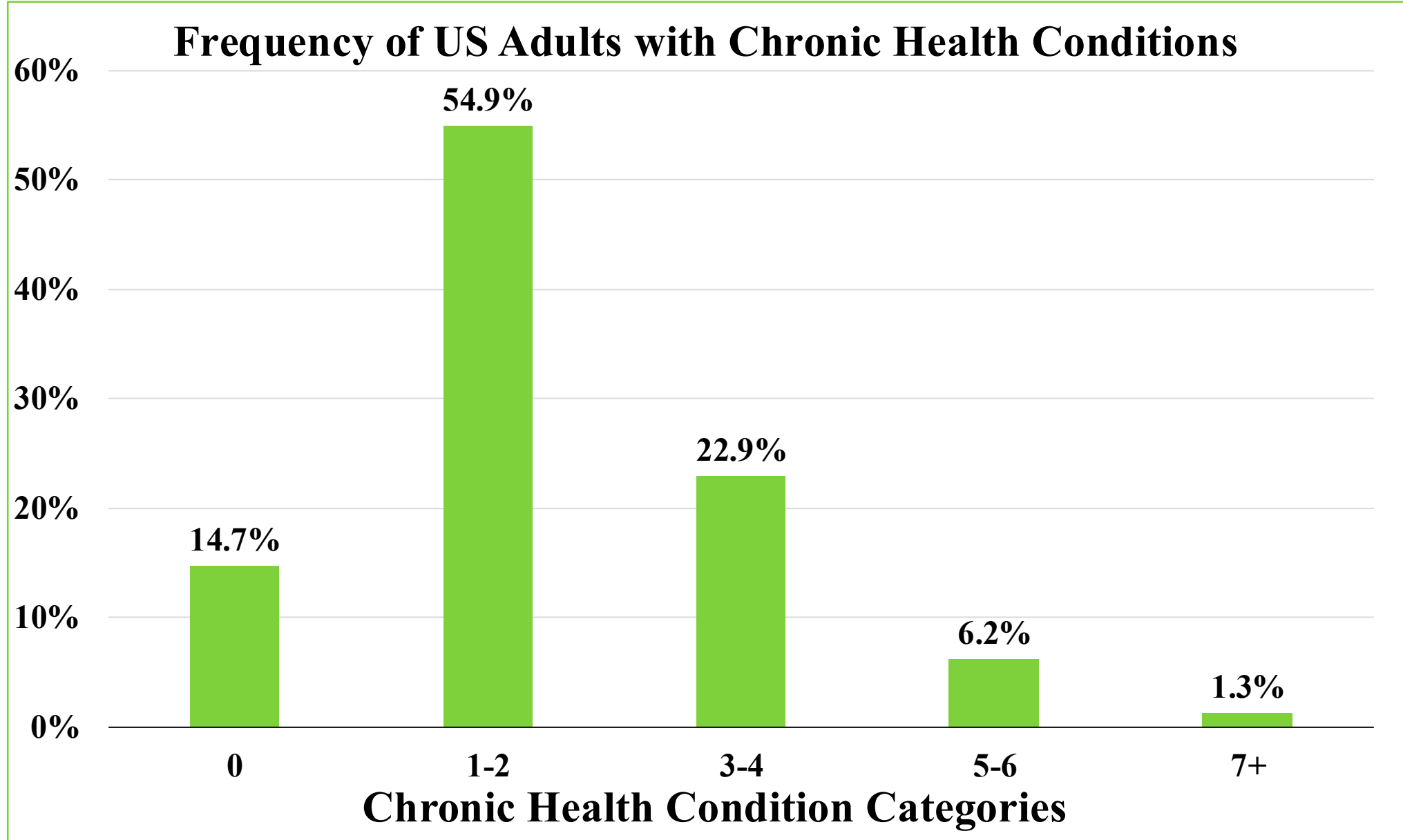
Bivariate analysis -> produced crude logistic regression models measuring the association between long-term COVID effects, chronic health conditions, and all additional predictors.

Multivariable analysis -> adjusted logistic regression models measuring the association between long-term COVID effects, chronic multimorbidity, and all additional predictors.

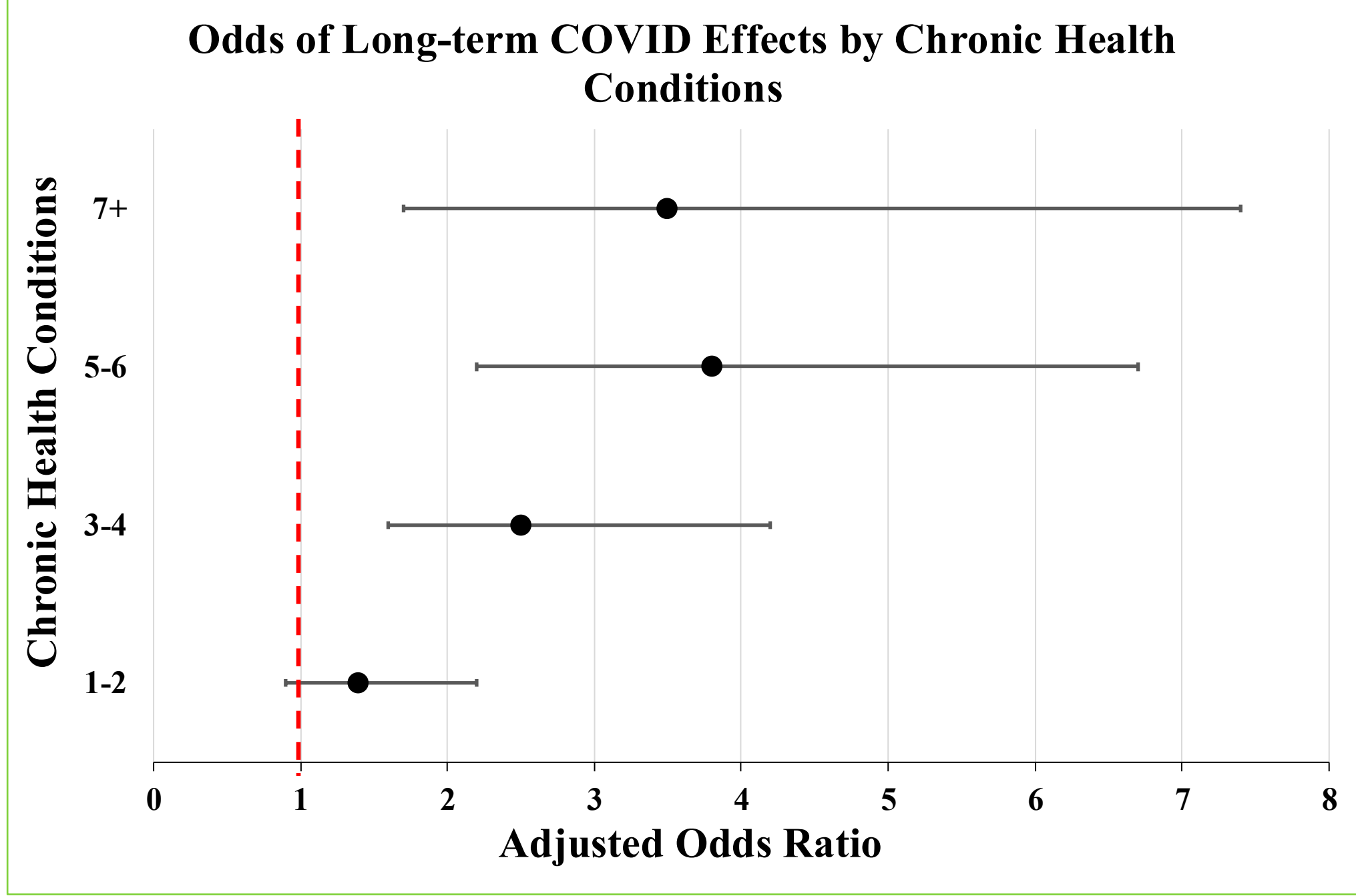
All analyses used the Survey PROCS in SAS to account for BRFSS's complex sample. This study used SAS version 9.4 for all analyses.

Results

Descriptive Analysis of Chronic Health Conditions



Predictor: Chronic Multimorbidity



- Overall, chronic multimorbidity was significantly associated with long-term COVID effects** ($p < .0001$). Compared to those with no chronic health conditions, those with 1-2 conditions had 1.4 (95% CI: 0.86 – 2.2) times greater odds of experience long-term COVID effects, those with 3-4 had 2.5 (95% CI: 1.6 – 4.2) times greater odds of experience long-term COVID effects, those with 5-6 had 3.8 (95% CI: 2.2 – 6.7) times greater odds of experience long-term COVID effects, and those with 7+ had 3.5 (95% CI: 1.7 – 7.4) times greater odds of experience long-term COVID effects.
- Sex, race/ethnicity, age, healthcare affordability, and chronic multimorbidity were significantly associated with long-term COVID effects.** Smoking status had a marginally significant association. Education, annual household income, health insurance status, and COVID vaccination status showed no significant association.

Proximal Risk Factors

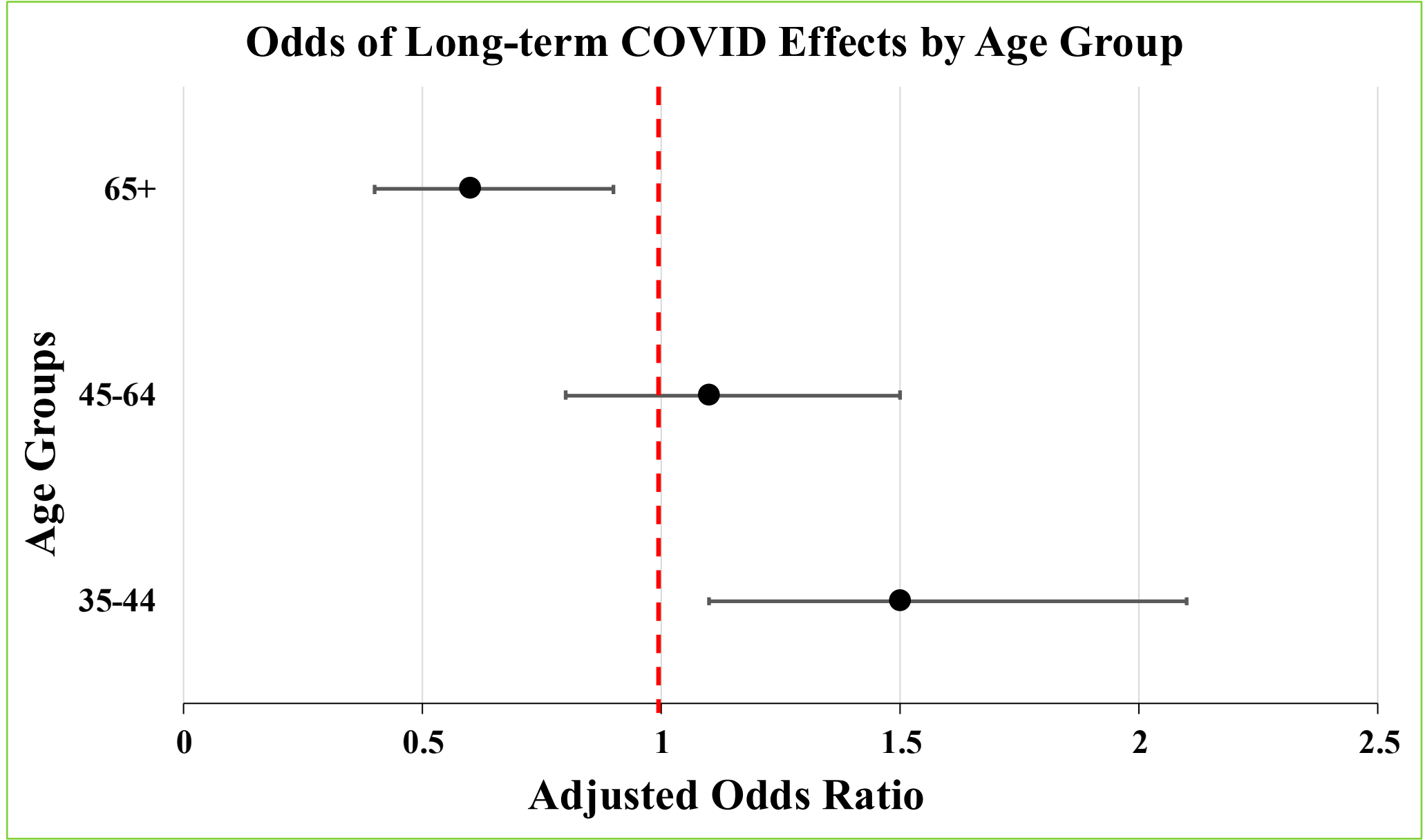
- Chronic Multimorbidity*** ($< .0001$)
- COVID vaccination status (0.2001)
- Smoking status*** (0.0531)

Distal Risk Factors

- Sex*** ($< .0001$)
- Race*** (.0095)
- Education (0.6380)
- Age*** ($< .0001$)
- Household annual income (0.1393)
- Health insurance coverage (0.7346)
- Healthcare affordability*** ($< .0001$)

Results

Predictor: Age

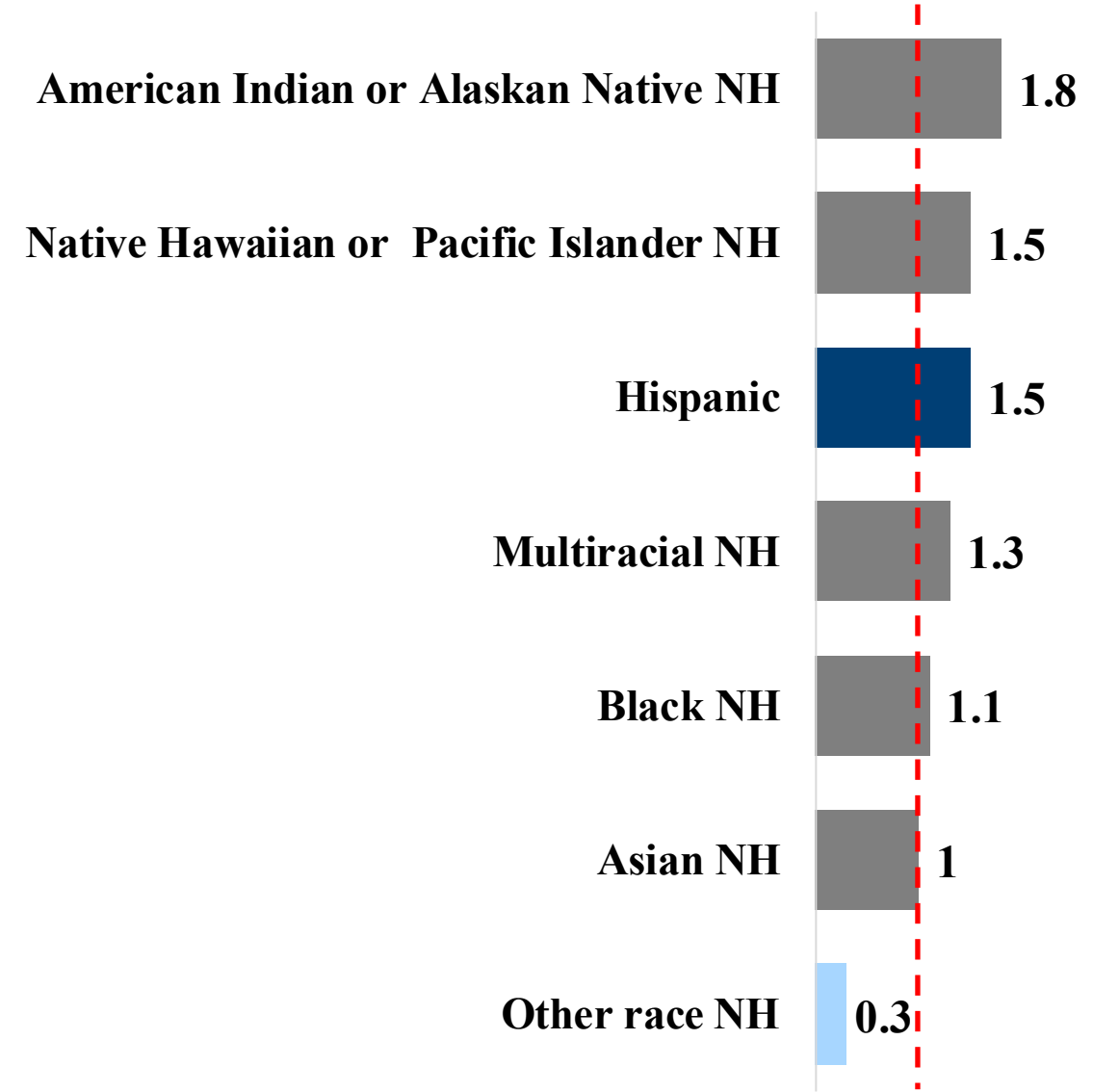


- Compared to 18-34-year-olds, the odds of experiencing long-term COVID effects got increasingly lower by increasing age groups. This could be a product of differences in social distancing and quarantine adherence between groups. Additionally, between 2020 and 2023, total deaths involving COVID increased with age. Older groups having higher death incidences lowers their chances of surviving to develop long-term effects of COVID.

Predictor: Race/Ethnicity

- Compared to the White non-Hispanic (NH) category, **Hispanic was the only racial/ethnic group that had significant greater odds, at 1.5 (95% CI: 1.1 – 2.1), of experiencing long-term COVID effects** ($p = 0.0088$). **"Other NH" was the only group that had significant lower odds, at 0.3 (95% CI: 0.11 – 0.74), of long-term COVID effects** ($p = 0.0093$).

Odds of Long-term COVID Effects by Racial and Ethnic Groups



Predictor: Health Care Affordability

Health care affordability was determined by an affirmative response to being able to afford a doctors visit in the past year. Compared to those who could afford a doctor's visit, **those who could not afford a doctor's visit had 1.9 (95% CI: 1.5 – 2.6) times greater odds of long-term COVID effects.**

Discussion + Implications + Future Research

- Research findings demonstrate a strong association between chronic multimorbidity and long-term COVID effects. More multimorbidity was associated with higher odds of long-term effects.
- SES related predictors had varying degrees of significance and effect. The **one significant SES related predictor, healthcare affordability, sits at the intersection of income and health care access.** Additionally, the insignificance of wealth indicators, like education, income, and health insurance access, point to issues with healthcare affordability that expands past low-income populations. Healthcare affordability presents as a fundamental factor because it works upstream to impact chronic multimorbidity, and directly—as demonstrated in this study—to influence long-term COVID effects.
- Future research should further evaluate this intersection and inform policy efforts to promote affordable healthcare.**